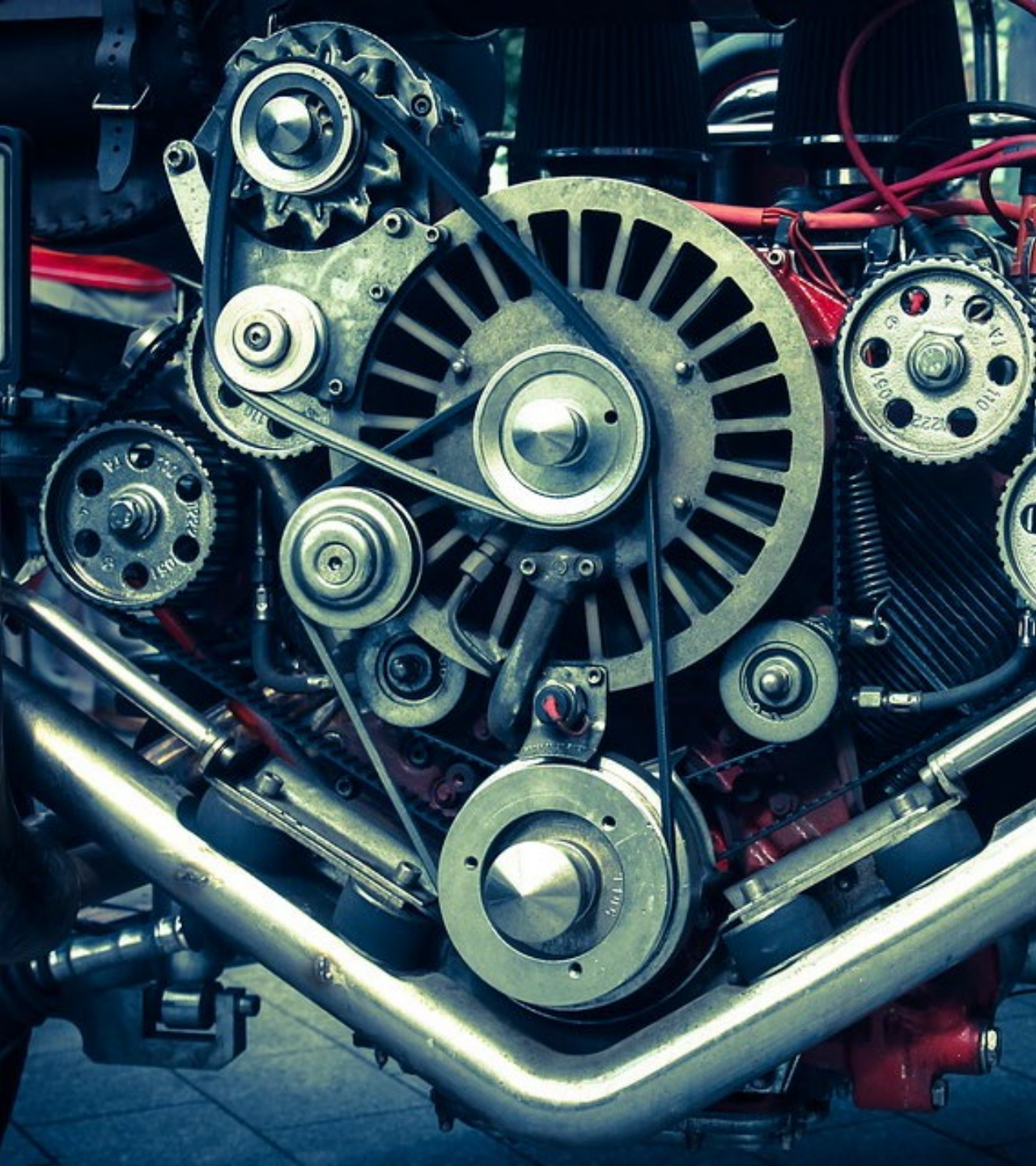
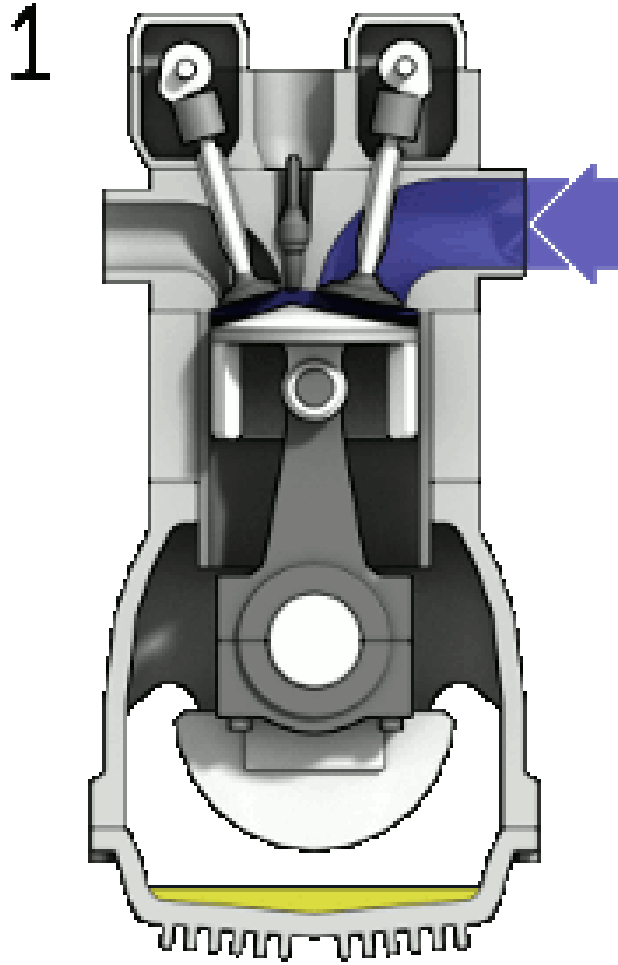




ME 300: Engineering Thermodynamics

Otto Cycle

Otto cycle



Ideal

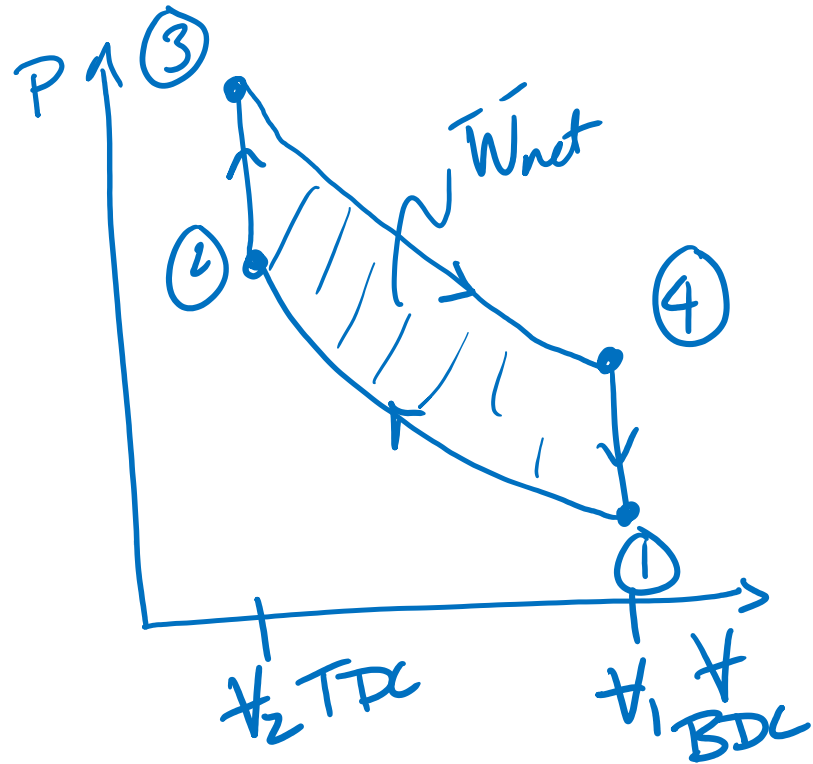
1-2: isentropic compression $\Delta S=0$

2-3: isochoric heat addition $\Delta V=0$

3-4: isentropic expansion $\Delta S=0$

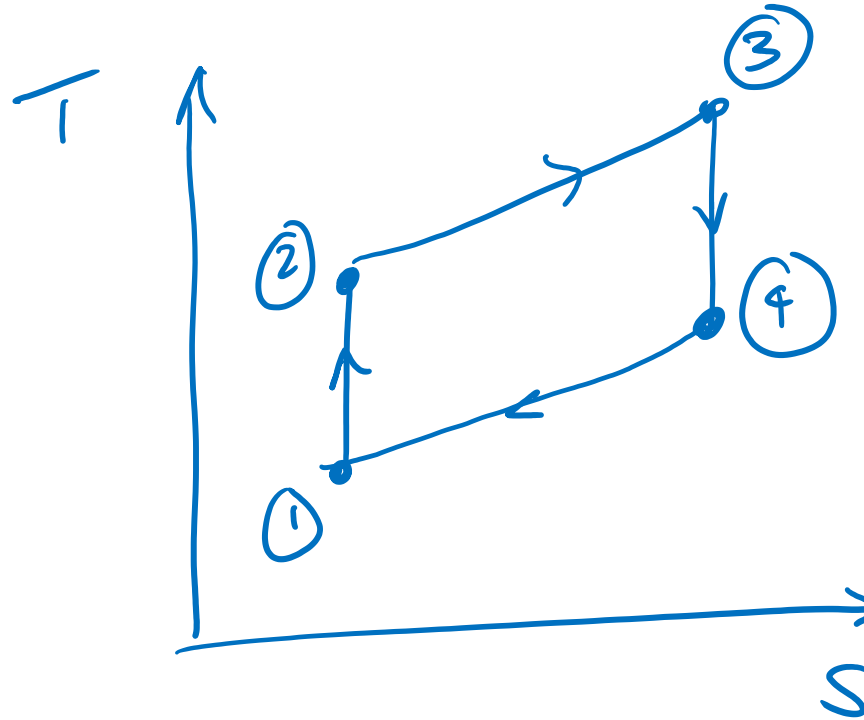
4-1: isochoric heat rejection

Otto cycle



$$r = \frac{v_1}{v_2}$$

$$d = v_1 - v_2$$



$$\bar{W}_{net} = {}_1\bar{W}_2 + {}_3\bar{W}_4$$

$$\eta_{otto} = \frac{{}_1\bar{W}_2 + {}_3\bar{W}_4}{{}_2Q_3}$$

Cycle analysis

$$\text{1st Law: } \Delta E = Q - W \rightarrow \Delta KE = \Delta PE = 0 \Rightarrow \Delta U = Q - W$$

$$1-2: \Delta U = -W \rightarrow {}_1\bar{W}_2 = -M C_V (T_2 - T_1)$$

$$2-3: \Delta U = Q \rightarrow {}_2Q_3 = M C_V (T_3 - T_2)$$

$$3-4: \Delta U = -W \rightarrow {}_3\bar{W}_4 = -M C_V (T_4 - T_3)$$

$$\eta_{th} = \frac{{}_1\bar{W}_2 + {}_3\bar{W}_4}{{}_2Q_3}$$

$$\eta_{th} = 1 - \frac{1}{r^{\gamma-1}}$$